

POWER BI

Interview Revision Notes

Complete Interview Preparation Guide — Concise & Print-Ready

SECTION 1 — POWER BI BASICS

1.1 What is Power BI?

Microsoft's business analytics tool that connects to multiple data sources, transforms raw data, and creates interactive dashboards and reports.

Example: A sales manager connects to Excel + SQL, builds a dashboard showing monthly revenue trends, and shares it with the team in one click.

Key Interview Points

- **Definition:** BI tool by Microsoft for data visualization and analysis
- **Main activities:** Connect → Transform → Visualize → Share
- Suitable for **both beginners and experts**
- **Language for calculations:** DAX (not SQL)

★ FREQUENTLY ASKED Q: Difference between Power BI and Tableau?

Feature	Power BI	Tableau
Language	DAX (measures)	MDX (measures & dimensions)
Data Handling	Limited data volume	Handles very large data
Ease of Use	Beginner + Expert friendly	Best suited for experts
UI	Simple, drag-and-drop	Complex but powerful
Cost	Free desktop version	Expensive licensing

★ FREQUENTLY ASKED Q: Difference between Power BI and Excel?

Feature	Power BI	Excel
Primary Use	Interactive dashboards & BI	Data analysis & calculations
Data Refresh	Automated refresh	Manual (unless VBA/Power Query)
Data Modeling	Star schema, DAX, relationships	Basic formulas & tables
Visualization	Advanced, interactive visuals	Basic charts, static graphs
Cross-Filtering	Advanced cross-filtering	Not available
Choose When	Dynamic reporting, multiple sources	Quick individual analysis

1.2 Power BI Data Sources

- Files: **.xlsx** (Excel), **.csv**, **.xml**, **.pbix**
- Databases: **SQL Server**, **Azure SQL**, **Azure HD Insight**
- Use **Get Data** feature to connect to any source

1.3 Power BI Desktop

Free tool to connect, transform, and visualize data. Updates monthly. Reports created here but sharing requires Pro license.

- **Free to use** — no cost for report creation
- **Drag-and-drop interface** — no coding needed
- Supports **complex DAX calculations**
- Works with Excel, SQL Server, Azure, and more

★ FREQUENTLY ASKED Q: What are the Views available in Power BI?

View	Purpose
Report View	Create and interact with visuals, charts, tables
Table View (Data View)	See raw data in rows/columns; sort, filter, format
Model View	Manage relationships between tables (one-to-many, many-to-many)
DAX View	Write DAX for custom measures, calculated columns, time functions

1.4 Building Blocks of Power BI

Block	What It Is
Dataset	Collection of data from sources like SQL, Azure, XML, JSON
Visualization	Charts, maps, tables that represent data visually
Report	Multi-page structured view of a dataset for detailed analysis
Dashboard	Single-page high-level view combining visuals from multiple reports

★ FREQUENTLY ASKED Q: Difference between Report and Dashboard?

Feature	Report	Dashboard
Pages	Multiple pages	Single page only
Data Source	One dataset per report	Multiple datasets/reports
Created In	Desktop & Service	Only in Power BI Service
Purpose	Detailed analysis	High-level summary
Use When	Deep-dive analysis needed	Quick overview required

1.5 Power BI Versions

Version	Cost	Key Feature
Desktop	Free	Create reports, cannot share without Pro
Pro	\$10/user/month	Share reports, collaborate, 8 refreshes/day
Premium Per User	\$20/user/month	Advanced features, 48 refreshes/day
Premium Capacity	\$4,995/month	Dedicated storage (50GB–100TB), free users can view
Service (Cloud)	Requires Pro/Premium	Web-based hosting, real-time collaboration

1.6 Filters in Power BI

★ FREQUENTLY ASKED Q: What are the 3 levels of filters?

Filter Level	Scope	Use Case
Visual Level	Only one specific visual	Filter one chart without affecting others
Page Level	All visuals on one page	Filter entire page without touching other pages
Report Level	All pages in report	Consistent filter across all report pages

1.7 Slicers in Power BI

Slicers are visual filters on the report canvas. They let end users filter data interactively.

Analogy: Think of a slicer as a dropdown menu in a website — you pick an option and the entire page updates.

Slicer Type	Best For
List Slicer	Checkbox-style filtering
Dropdown Slicer	Saves space, collapsible
Date Slicer	Filter by date range
Numeric Range Slicer	Filter numbers within a range
Hierarchy Slicer	Drill down: Region → Country → City
Relative Date Slicer	Dynamic: last 30 days, next quarter

1.8 Visualizations in Power BI

★ FREQUENTLY ASKED Q: What chart types are available in Power BI?

Category	Chart Types	Best Used For
Bar/Column	Stacked, Clustered, 100% Stacked	Compare categories, show proportions
Line/Area	Line, Area, Stacked Area	Trends over time
Pie/Donut	Pie, Donut	% share (max 5-6 categories only)

Table/Matrix	Table, Matrix (like Pivot)	Raw data, hierarchical analysis
Cards/KPI	Card, Multi-row Card, KPI	Single key metrics, goal tracking
Scatter/Bubble	Scatter, Bubble	Relationship between two variables
Interview Tip: KPI has 3 parts — Indicator (actual value), Target (goal), Trend (arrow/color showing direction)		

1.9 Custom Visuals

Type	Description
Certified Custom Visuals	Approved by Microsoft. Safe to use. Example: Gantt Chart, Bullet Chart
Non-Certified Custom Visuals	Third-party. Not Microsoft-approved. Cannot export. Example: Network Graph
Custom Developed Visuals	Built using Power BI Visuals SDK. For unique business needs
Interview Tip: Always use Microsoft CERTIFIED custom visuals in production for security and performance.	

1.10 Content Pack

Pre-built ready-to-use dashboards/reports created by Microsoft or 3rd parties (e.g., Google Analytics, Salesforce). Saves time — no need to build from scratch.

1.11 Group By

Groups rows by one or more columns and calculates aggregates (SUM, AVG, COUNT, MAX, MIN). Same concept as SQL GROUP BY.

Example: Group sales by Region → get Total Sales per region.

1.12 Data Storage in Power BI

- **Azure Blob Storage** — stores uploaded user data
- **Azure SQL Database** — stores all metadata, fact tables, and dimension tables

1.13 Comprehensive Working System

Step	Action
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1. Data Integration	Extract and connect data from multiple/different sources
2. Data Processing	Clean, transform, and load data to data warehouse
3. Data Presentation	Display cleaned data as reports, dashboards, scorecards

1.14 Performance — Common Mistakes & Fixes

★ FREQUENTLY ASKED Q: How do you optimize slow Power BI reports?

Mistake	Fix
Importing unnecessary columns/rows	Use Power Query to reduce columns & rows before loading
Complex/nested DAX formulas	Use DAX measures (not calculated columns), optimize formulas
Too many visuals on one page	Limit visuals per page for fast rendering
Using Direct Query unnecessarily	Use Import Mode when real-time data is not required
No data modeling standard	Follow Star Schema for better performance

SECTION 1 REVISION — Power BI Basics □ Power BI: Connect → Transform → Visualize → Share □
 Report = multi-page detail | Dashboard = single-page summary □ 3 Filter levels: Visual → Page →
 Report □ Import Mode = faster | Direct Query = real-time □ Always use Star Schema for best
 performance □ KPI = Indicator + Target + Trend

SECTION 2 — POWER QUERY (ETL)

2.1 What is Power Query?

Power Query is the ETL (Extract, Transform, Load) tool in Power BI. It imports data from sources, cleans it, and prepares it for analysis.

Analogy: Power Query is like a car wash — raw dirty data goes in, clean data comes out.

- Uses **M Query Language** for calculations
- You write the query **once**, run it with a simple refresh
- Original data source remains **unchanged**

2.2 4 Phases of Power Query

Phase	Action
1. Connect	Import and connect to cloud/local data sources
2. Transform	Shape data to meet needs — original source unchanged
3. Combine	Merge data from multiple sources into one unified view
4. Load	Save query to worksheet or data model

2.3 Common Power Query Tasks

- Connect to data | Group rows | Create custom columns
- Shape and combine data | Pivot and unpivot columns
- **Filling Down or Up** | Query formulas | Filter/remove rows
- **Transformations:** Text, Number, Date transformations, conditional columns, replace values

★ FREQUENTLY ASKED Q: Difference between Append and Merge in Power Query?

Operation	Equivalent to	Use When
Append	UNION in SQL	Combine 2 tables row by row (same structure)
Merge	JOIN in SQL	Combine 2 tables based on a matching column/key

★ FREQUENTLY ASKED Q: What is Query Folding?

When Power BI translates M Query transformations into native SQL queries and sends them to the database — instead of pulling all data and transforming in Power BI.

Analogy: Instead of ordering a huge raw meal and cooking it yourself, you tell the restaurant exactly what you want — query folding tells the database to do the work.

- Converts: **M Query** → **SQL** → **sent to DB** → **result returned**
- When it occurs: Filtering rows, removing columns, renaming, grouping, merging, aggregations
- **Benefit:** Faster performance, less memory usage

★ FREQUENTLY ASKED Q: Difference between Power Query and DAX?

Parameter	Power Query	DAX
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Purpose	Data preparation & ETL (cleaning)	Data modeling & calculations
Language	M Language (functional)	DAX (similar to Excel formulas)
When it runs	During data load/refresh	After load — during reporting & visualization
Key functions	Filter, merge, unpivot, clean	Measures, calculated columns, time intelligence
Use for	Clean raw data, combine sources	SUM, AVERAGE, CALCULATE, time analysis

2.4 Power Pivot

Data modeling tool that creates relationships between tables and builds calculated measures using DAX (after Power Query cleans the data).

Power Query	Power Pivot
Clean/reshape data	Model and analyze data
Uses M function	Uses DAX function
More memory (M)	Less memory (DAX)
ETL Stage	Modeling Stage
Memory Trick: Power Query = COOK the data. Power Pivot = SERVE and analyze the data.	

SECTION 2 REVISION — Power Query □ Power Query = ETL tool using M Language □ Append = stacking tables (like UNION in SQL) □ Merge = joining tables (like JOIN in SQL) □ Query Folding = Power BI sends work to database instead of doing it itself □ Power Query cleans data; Power Pivot models data

SECTION 3 — DATA MODELING

★ FREQUENTLY ASKED Q: What are Fact Tables and Dimension Tables?

3.1 Fact Table vs Dimension Table

Imagine a grocery store database. The Sales Table (what was sold, when, by whom) is the Fact Table. Customer details, Product details are Dimension Tables (lookup data).

Type	Contains	Example
Fact Table	Transactional/measurable data (numbers)	Sales: Date, Item_ID, Customer_ID, Order_Weight
Dimension Table	Descriptive/lookup data	Customer: ID, Name, Email, Address

- Fact table has **foreign keys** linking to dimension tables
- Fact table is **highly normalized** — no repeated details
- Dimension tables prevent **data redundancy** (change address once, not 100 rows)

★ FREQUENTLY ASKED Q: What is Star Schema vs Snowflake Schema?

3.2 Star Schema vs Snowflake Schema

Feature	Star Schema	Snowflake Schema
Structure	Fact table in center, dimension tables around it	Dimension tables connected to other dimension tables
Shape	Looks like a star ★	Looks like a snowflake ❄️
Joins	Simple joins	Complex joins
Performance	Better (faster queries)	Slower (more join operations)
Recommended	Yes — Power BI best practice	Avoid if possible in Power BI
Memory Trick: Star = Simple (one level of dimension tables). Snowflake = Complex (dimension tables branching further).		

3.3 Import Mode vs Direct Query Mode

★ FREQUENTLY ASKED Q: When do you use Import Mode vs Direct Query?

Feature	Import Mode	Direct Query Mode
Data Storage	Stored in Power BI (in-memory)	Stays in the source database
Performance	Faster (in-memory processing)	Slower (DB queried every interaction)
Data Refresh	Manual/scheduled refresh	Real-time (always live)
File Size	Larger (.pbix file)	Smaller (no data stored)
DAX Support	Fully supported	Limited DAX

Best For	Small-medium datasets, fast reports	Large datasets, real-time dashboards
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3.4 Bidirectional Cross-Filtering

Normally filters flow in one direction (from dimension to fact table). Bidirectional lets filters flow BOTH ways between two tables.

Analogy: Normal filtering is a one-way street. Bidirectional is a two-way street — both tables can filter each other.

- **Use when:** Many-to-many relationships, complex data models
- **Avoid when:** Large datasets (slower), risk of circular loops
- **Alternative:** Use `crossfilter()` function in DAX

3.5 Data Modeling — Desktop vs Excel Power Pivot

Feature	Power BI Desktop	Excel Power Pivot
Relationships	Bi-directional cross-filtering	Single direction only (one-to-many)
Calculated Tables	Supported	Not supported
Import Options	Multiple import modes	Import mode only
Security (RLS)	Supported	Not available

SECTION 3 REVISION — Data Modeling

- Fact Table = transaction data | Dimension Table = lookup/descriptive data
- Star Schema = best practice (simple joins, better performance)
- Snowflake Schema = avoid in Power BI (complex joins)
- Import Mode = fast, needs refresh | Direct Query = live data, slower
- Bidirectional filtering = both tables filter each other

SECTION 4 — POWER BI SERVICE

4.1 My Workspace vs Workspace

Feature	My Workspace	Workspace
Purpose	Personal use only	Team collaboration
Access	Only you can see/edit	Multiple team members
Analogy	Your personal rough notebook	Team's shared class notebook
Use When	Testing, personal reports	Project-wide shared reports

★ FREQUENTLY ASKED Q: What are the workspace roles in Power BI?

4.2 Workspace Roles

Role	Can Do	Best For
Admin	Full control — add/remove users, manage all permissions, delete content, publish	IT Admins, Managers
Member	Create, edit, publish content. Cannot manage user permissions	Report Developers
Contributor	Add and edit content. Cannot publish or manage permissions	Analysts
Viewer	Read-only. View reports and dashboards only	End Users, Stakeholders
Memory Trick: Admin > Member > Contributor > Viewer (AMCV — Always Make Clear Visuals!)		

★ FREQUENTLY ASKED Q: What is RLS (Row Level Security)?

4.3 Row Level Security (RLS)

Restricts which data rows a specific user can see in a report. Different users see different slices of the same report.

Example: India team sees only India sales data. US team sees only US data — same dashboard, different data.

Type	How It Works	Drawback
Static RLS	Manually assign users to roles/regions in Manage Rules	Manual — must add people one by one; done in Desktop, not Service
Dynamic RLS	Uses USERPRINCIPALNAME() DAX function to auto-detect logged-in user	Needs an entitlement/mapping table setup

- **DAX Function:** USERPRINCIPALNAME() — captures the logged-in user's email
- **Setup:** Define roles in Desktop → publish to Service → assign users in Service

★ FREQUENTLY ASKED Q: What is a Gateway in Power BI?

4.4 Gateway

A secure bridge between on-premises data (local SQL, Excel, SharePoint) and Power BI Service (cloud). Without it, cloud can't access local data.

Analogy: Gateway is like a postman — it picks up data from your local office and delivers it to the cloud.

Feature	Personal Gateway	Standard (Enterprise) Gateway
Users	Single user only	Multiple users
DirectQuery	Not supported	Supported
Refresh Type	Scheduled refresh only	Both DirectQuery & Scheduled
Services	Power BI only	Power BI, PowerApps, Power Automate, Azure Logic Apps
Best For	Personal practice	Enterprise/organization level

4.5 Dataflow in Power BI Service

Dataflow = Power Query Editor running inside Power BI Service (cloud). You can create, transform, and reuse datasets across different workspaces without going back to Desktop.

One-liner: Dataflow is Power Query's twin that lives in the cloud.

4.6 Refresh Options

Refresh Type	Description	Best For
Scheduled Refresh	Define frequency & time slots (8/day for Pro, 48/day for Premium)	Daily sales reports, periodic updates
Incremental Refresh	Only refreshes new/changed data, not entire dataset	Large datasets where recent data changes frequently

4.7 Self-Service BI (SSBI)

Enables non-technical business users (no coding background) to build reports and draw insights from data independently using Power BI's visual interface.

4.8 Bookmarks in Power BI

A bookmark saves the state of a report page (filters, slicers, visuals). Users can navigate between different views without manually resetting filters.

- **Used for:** Navigation buttons, storytelling, dynamic reports
 - Combine bookmarks + **buttons + actions** to make reports interactive/dynamic
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4.9 Custom Visual Files

Use custom visual files (.pbviz) when built-in visuals don't meet business needs. Import like any pre-packaged visual.

- **Always use Microsoft-certified custom visuals** in production reports
- **Source:** Get from AppSource for trusted third-party visuals

4.10 General Formatting Best Practices

Element	Best Practice
Fonts	Use clear fonts like Segoe UI
Colors	Consistent brand colors, good contrast for accessibility
Numbers	Thousand separators, currency symbols, decimal precision
Dates	Consistent date formats across all visuals
Titles	Meaningful titles and axis labels on every chart
KPIs	Place high-level KPIs at top of the report
Tooltips	Add insightful tooltips for detailed hover context

SECTION 4 REVISION — Power BI Service □ Admin > Member > Contributor > Viewer (AMCV) □ RLS = users see only their data; Dynamic RLS uses USERPRINCIPALNAME() □ Gateway = bridge between local data and cloud □ Personal Gateway = single user | Standard Gateway = team/enterprise □ Dataflow = Power Query inside cloud □ Bookmark = saves report state for navigation □ Incremental Refresh = only refreshes new/changed data

SECTION 5 — DAX (Data Analysis Expressions)

5.1 What is DAX?

Collection of functions, operators, and constants used to write formulas that calculate and return values in Power BI. It creates new insights from existing data.

- Runs **after data is loaded** (during reporting and visualization)
- Used for: custom aggregations, relationships, filtering, time-based analysis

★ FREQUENTLY ASKED Q: What are the 3 fundamental concepts of DAX?

5.2 Three Fundamental DAX Concepts

Concept	Description	Example
1. Syntax	How formulas are written. Syntax error = error message.	Total Sales = SUM(Sales[Revenue])
2. Functions	Predefined formulas (Aggregation, Logical, Filter, Time Intelligence)	SUM(), IF(), CALCULATE(), TOTALYTD()
3. Context	Determines which data is evaluated (Row Context vs Filter Context)	Row Context: calculated columns Filter Context: measures

★ FREQUENTLY ASKED Q: Row Context vs Filter Context?

5.3 Row Context vs Filter Context

Aspect	Row Context	Filter Context
Definition	Current row being calculated	Subset of data visible based on applied filters
Where used	Calculated columns, iterators (SUMX, FILTER)	Measures, slicers, visual filters, CALCULATE()
Scope	One row at a time	Entire dataset (filtered)
Example	TotalPrice = Sales[Qty] * Sales[UnitPrice]	TotalSales = SUM(Sales[Revenue]) — changes with slicer

★ FREQUENTLY ASKED Q: Measure vs Calculated Column?

5.4 Measure vs Calculated Column

Feature	Measure	Calculated Column
Calculation	Aggregated (dynamic)	Row-by-row (static)
Storage	NOT stored, calculated on demand	Stored in data model (increases size)
Performance	More efficient	Slower (increases model size)
Context	Changes based on filters/slicers	Fixed at data load time

Example	Total Sales = SUM(Sales[Revenue])	Profit Margin = Sales[Profit] / Sales[Revenue]
Use When	SUM, AVERAGE, ratios, aggregations	Row-level: concatenations, categorizations
Memory Trick: MEASURE = dynamic, not stored (like a live calculation). COLUMN = static, stored (like a new column you added to the table).		

5.5 DAX Function Categories

Category	Functions	Use For
Aggregation	SUM(), AVERAGE(), MIN(), MAX(), COUNT()	Summarize data
Logical	IF(), AND(), OR(), SWITCH()	Test conditions, return TRUE/FALSE
Filter	FILTER(), CALCULATE(), ALL(), RELATED()	Control filter context
Time Intelligence	TOTALYTD(), SAMEPERIODLASTYEAR(), DATEADD()	Date-based calculations
Text	CONCATENATE(), FORMAT(), LEFT(), RIGHT(), MID(), LEN(), TRIM()	Text manipulation
Mathematical	ABS(), POWER(), MOD(), SUM(), AVG()	Math operations

★ FREQUENTLY ASKED Q: Difference between SUM() and SUMX()?

5.6 SUM() vs SUMX()

Feature	SUM()	SUMX()
What it does	Adds all values in a single column	Iterates over rows, evaluates expression, then sums
Syntax	SUM(<column>)	SUMX(<table>, <expression>)
Row-by-row calc	No	Yes
Performance	Faster	Slightly slower (row iteration)
Use When	Simple total from one column	Need to calculate per row then sum (e.g., Qty * Price)
Example	Total Sales = SUM(Sales[Revenue])	Total Revenue = SUMX(Sales, Sales[Qty] * Sales[Price])

5.7 CALCULATE() vs CALCULATETABLE()

Function	Returns	Example
CALCULATE()	A single scalar value	Total_Electronics = CALCULATE(SUM(Sales[Amount]), Products[Category]="Electronics")
CALCULATETABLE()	A full table (filtered)	Electronic_Sales = CALCULATETABLE(Sales, Products[Category]="Electronics")

5.8 ALL() vs ALLEXCEPT()

Function	Effect	Use Case
ALL()	Removes ALL filters from a table/column	Calculate grand total ignoring slicers
ALLEXCEPT()	Removes all filters EXCEPT specified columns	Keep grouping filters, remove others

★ FREQUENTLY ASKED Q: What is SAMEPERIODLASTYEAR()?

5.9 SAMEPERIODLASTYEAR()

Compares current period with the SAME period last year (not the previous month).

Example: Compare January 2025 sales with January 2024 (not December 2024 — seasonality makes that comparison misleading).

5.10 SUMMARIZE() Function

Creates a new summary table grouped by specified columns with aggregations. Works like SQL GROUP BY.

- Syntax: SUMMARIZE(table, groupBy_column, "NewColName", expression)
- Example: SUMMARIZE(Sales, Product[Category], "Total Sales", SUM(Sales[Revenue]))

5.11 Date Master Table

A dedicated date dimension table used as a reference for all time intelligence functions. Connect all fact tables to this single date table.

- **Purpose:** Acts as a single reliable date reference for TOTALYTD, SAMEPERIODLASTYEAR, etc.
- Create with CALENDAR() or CALENDARAUTO() DAX functions

5.12 DISTINCT() vs VALUES()

Function	Returns	Key Difference
DISTINCT()	Unique values only (no blank rows)	Ignores blank/null rows
VALUES()	Unique values + includes blank row if it exists	Respects current filter context

5.13 MAX() vs MAXA() vs MAXX()

Function	Works On	Handles Logical?
MAX()	Numeric, date, text columns	No — ignores TRUE/FALSE
MAXA()	Same + Logical values (TRUE=1, FALSE=0)	Yes — TRUE/FALSE become 1/0
MAXX()	Iterates over table rows, evaluates expression per row	Yes — returns max of calculated expression

5.14 IF() vs SWITCH()

Feature	IF()	SWITCH()
Purpose	Simple true/false logic	Multiple conditional checks
Nesting	Requires nested IFs for complexity	Handles multiple conditions without nesting
Best Use	Binary classification (yes/no)	Category mapping (High/Medium/Low)
Tip	Use AND()/OR() for complex logic	Use SWITCH(TRUE(), ...) for conditional ranges

5.15 DATEDIFF() vs DATESINPERIOD()

Function	Returns	Syntax	Use For
DATEDIFF()	Number (difference between 2 dates)	DATEDIFF(start, end, interval)	Calculate delivery days, age, tenure
DATESINPERIOD()	Table of dates in a range	DATESINPERIOD(dates, start, intervals, interval)	Get last 3 months, next quarter dates

- Positive intervals = look forward | Negative intervals = look backward

5.16 Key TEXT() Functions

Function	Purpose	Example
CONCATENATE()	Join two text strings	CONCATENATE(FirstName, LastName) → "John Doe"
FORMAT()	Convert to formatted text	FORMAT(TODAY(), "MMMM YYYY") → "February 2025"
LEFT() / RIGHT()	Extract N chars from start/end	LEFT("PowerBI", 5) → "Power"
MID()	Extract substring from position	MID("PowerBI", 3, 3) → "wer"
LEN()	Count characters	LEN("PowerBI") → 7
TRIM()	Remove extra spaces	TRIM(" Power BI ") → "Power BI"
UPPER/LOWER/PROPER()	Change case	UPPER("power bi") → "POWER BI"

FIND() / SEARCH()	Find position of text	SEARCH = case-insensitive; FIND = case-sensitive
REPLACE() / SUBSTITUTE()	Replace text	REPLACE: by position; SUBSTITUTE: by value

SECTION 5 REVISION — DAX 3 DAX Concepts: Syntax, Functions, Context

- Row Context = calculated columns
- Filter Context = measures/slicers
- Measure = dynamic, not stored
- Calculated Column = static, stored
- SUM() = simple column total
- SUMX() = row-by-row then sum
- CALCULATE() = returns a value
- CALCULATETABLE() = returns a table
- ALL() removes all filters
- ALLEXCEPT() removes all except specified
- SAMEPERIODLASTYEAR() = compare same month, not previous month

SECTION 6 — ADDITIONAL IMPORTANT TOPICS (Not in PDF)

6.1 Power BI Architecture

Component	Role
Power BI Desktop	Create reports (free tool)
Power BI Service	Publish, share, collaborate (cloud)
Power BI Mobile	View reports on iOS/Android
Power BI Gateway	Connect on-prem data to cloud
Power BI Embedded	Embed reports into custom applications (for developers)
Power BI Report Server	On-premises report hosting (for orgs that can't use cloud)

6.2 Relationships in Power BI

Type	Description	Example
One-to-One (1:1)	Each row in Table A matches exactly one row in Table B	Employee ↔ EmployeeDetails
One-to-Many (1:*)	One row in A matches many rows in B (most common)	Customer ↔ Orders
Many-to-Many (*:*)	Multiple rows in A match multiple rows in B (use carefully)	Students ↔ Courses

- **Most common in Star Schema:** One-to-Many (dimension → fact table)
- **Cardinality:** The type of relationship between tables

6.3 Drill Down vs Drill Through

Feature	Drill Down	Drill Through
Direction	Goes deeper into same visual	Navigates to a different detail report page
Example	Year → Quarter → Month → Day in a chart	Click on a product → go to product detail page
How to use	Hierarchy in visual	Set up a drill-through page with filters

6.4 Power BI Q&A (Natural Language Query)

Allows users to type questions in plain English and get instant visual answers. Uses AI to interpret questions. Example: Type 'What were last month's total sales?' → Power BI shows an instant chart.

6.5 Aggregation Types in Power BI Visuals

- **SUM** — total of values | **AVERAGE** — mean | **MIN/MAX** — smallest/largest
- **COUNT** — count of rows | **COUNT DISTINCT** — count unique values

- **Do Not Summarize** — shows raw values without aggregation

6.6 Power BI File Types

Extension	Description
.pbix	Power BI Desktop file (data + report + model)
.pbix	Power BI Template (no data, just structure)
.pbiviz	Custom visual file
.csv / .xlsx	Common data source file types

6.7 RELATED() vs LOOKUPVALUE()

Function	Use Case	Requires
RELATED()	Pull value from related table (via relationship)	An existing relationship between tables
LOOKUPVALUE()	Pull value from any table by matching conditions	No relationship needed — like VLOOKUP

6.8 RANKX() Function

Ranks items in a table based on a measure or expression.

- Syntax: RANKX(table, expression, [value], [order], [ties])
- Example: RANKX(ALL(Products), [Total Sales]) → ranks all products by sales

SECTION 6 BONUS TIPS □ Most common relationship: One-to-Many (dimension → fact) □ Drill Down = deeper in same visual | Drill Through = navigate to another page □ RELATED() needs relationship | LOOKUPVALUE() works like VLOOKUP (no relationship needed) □ .pbix = full file | .pbix = template (no data) | .pbiviz = custom visual

SECTION 7 — TOP INTERVIEW QUESTIONS & ANSWERS

Most Frequently Asked Questions

Question	One-Line Answer
What is Power BI?	Microsoft BI tool to connect, transform, visualize, and share data
Power BI vs Tableau?	Power BI = beginner-friendly, DAX, limited data. Tableau = experts, MDX, handles large data
Power Query vs DAX?	Power Query = clean data (M language). DAX = calculate after loading
Import Mode vs DirectQuery?	Import = fast, needs refresh. DirectQuery = live, slower performance
Measure vs Calculated Column?	Measure = dynamic aggregate (not stored). Calculated Column = static row-level (stored)
Star vs Snowflake Schema?	Star = simple joins, better performance. Snowflake = complex, slower
What is RLS?	Restricts rows users can see. Static = manual. Dynamic = uses USERPRINCIPALNAME()
What is a Gateway?	Bridge between on-premises data and Power BI cloud service
Report vs Dashboard?	Report = multi-page detail. Dashboard = single-page summary
SUM() vs SUMX()?	SUM() = simple column total. SUMX() = calculate per row, then sum
Row Context vs Filter Context?	Row Context = current row (calculated columns). Filter Context = data visible with filters (measures)
What is Query Folding?	Power BI converts M transformations to SQL and sends to DB for faster processing
What is CALCULATE()?	Modifies filter context and returns a single value
What are slicer types?	List, Dropdown, Date, Numeric Range, Hierarchy, Relative Date
3 filter levels?	Visual Level, Page Level, Report Level

Scenario-Based Questions (Common in Interviews)

Scenario	Best Answer
Report is loading slowly	Use Import mode, reduce columns in Power Query, limit visuals per page, use DAX measures not calculated columns, follow Star Schema
Need real-time data dashboard	Use DirectQuery mode, Standard Gateway for on-prem sources
Different users see different data	Implement Row Level Security (RLS) — Static for small teams, Dynamic (USERPRINCIPALNAME()) for large teams
Need to compare this year vs last year	Use SAMEPERIODLASTYEAR() time intelligence function

Need total ignoring slicer filters	Use CALCULATE(SUM(...), ALL(TableName))
Need month-over-month comparison	Use SAMEPERIODLASTYEAR() NOT previous month — seasonality differs
Data not updating automatically	Set up Scheduled Refresh in Power BI Service settings

1-DAY POWER BI REVISION SHEET

Print this page — Everything you need to revise before the interview!

POWER BI BASICS — Quick Recall

- ❑ Power BI workflow: Connect → Transform (Power Query) → Model (Power Pivot) → Visualize → Share
- ❑ Report = multiple pages, one dataset | Dashboard = single page, multiple reports/datasets
- ❑ 3 Filter levels: Visual → Page → Report (gets broader each level)
- ❑ 4 Views: Report, Table (Data), Model, DAX
- ❑ 4 Building Blocks: Dataset, Visualization, Report, Dashboard
- ❑ Import Mode = fast (needs refresh) | DirectQuery = live (slower)
- ❑ Workspace Roles: Admin > Member > Contributor > Viewer

POWER QUERY — Quick Recall

- ❑ Power Query = ETL tool using M Language — cleans and reshapes data
- ❑ Append = UNION (stack rows) | Merge = JOIN (combine on matching key)
- ❑ Query Folding = M Query → SQL → sent to DB (faster performance)
- ❑ 4 Phases: Connect → Transform → Combine → Load

DATA MODELING — Quick Recall

- ❑ Fact Table = transactions | Dimension Table = lookup/descriptive data
- ❑ Star Schema = BEST PRACTICE (fact in center, dimensions around, simple joins)
- ❑ Snowflake Schema = AVOID (dimension tables branch further, complex joins)
- ❑ Bidirectional filtering = both tables filter each other (use carefully — can slow performance)

POWER BI SERVICE — Quick Recall

- ❑ Gateway = bridge between local data and Power BI cloud
- ❑ Personal Gateway = single user | Standard Gateway = team (supports DirectQuery)
- ❑ RLS = restrict data by user | Dynamic RLS uses USERPRINCIPALNAME()
- ❑ Dataflow = Power Query inside Power BI Service (cloud)
- ❑ Bookmark = saves report state (filters + slicers + visuals) for navigation

DAX — Quick Recall

- ❑ 3 DAX Concepts: Syntax | Functions | Context
- ❑ Row Context = calculated columns (one row at a time)
- ❑ Filter Context = measures (filters from slicers, CALCULATE, visuals)
- ❑ Measure = dynamic, NOT stored | Calculated Column = static, STORED in model
- ❑ SUM() = simple column | SUMX() = iterate rows then sum
- ❑ CALCULATE() = scalar value | CALCULATETABLE() = returns a table
- ❑ ALL() = remove all filters | ALLEXCEPT() = remove all except specified
- ❑ SAMEPERIODLASTYEAR() = compare same month last year (NOT previous month)
- ❑ DISTINCT() = unique values, no blanks | VALUES() = unique + blank row if exists
- ❑ IF() = binary | SWITCH() = multiple conditions (cleaner for 3+ conditions)
- ❑ RELATED() = needs relationship | LOOKUPVALUE() = like VLOOKUP, no relationship needed

MUST-KNOW DAX FUNCTIONS CHEAT SHEET

Function	What it does	Quick Example
SUM()	Total of a column	SUM(Sales[Revenue])
SUMX()	Row-by-row calc then sum	SUMX(Sales, [Qty]*[Price])
CALCULATE()	Modify filter context	CALCULATE(SUM(Sales[Rev]), Year=2024)
FILTER()	Return filtered table	FILTER(Sales, Sales[Rev]>1000)
ALL()	Remove all filters	CALCULATE([Total Sales], ALL(Sales))
RELATED()	Get value from related table	RELATED(Customer[City])
IF()	True/false condition	IF([Sales]>1000, "High", "Low")
SWITCH()	Multiple conditions	SWITCH(TRUE(), [Rev]>=3000, "Premium", "Low")
TOTALYTD()	Year-to-date total	TOTALYTD(SUM([Rev]), Date[Date])
SAMEPERIODLASTYEAR() ()	Same period last year	CALCULATE([Sales], SAMEPERIODLASTYEAR(Date[Date]))
DISTINCTCOUNT()	Count unique values	DISTINCTCOUNT(Orders[CustomerID])
RANKX()	Rank items	RANKX(ALL(Products), [Total Sales])
USERPRINCIPALNAME()	Get logged-in user email	Used in Dynamic RLS

MEMORY TRICKS SUMMARY

Concept	Memory Trick
Power Query vs Power Pivot	Query = COOK the data Pivot = SERVE and analyze
Measure vs Calculated Column	Measure = LIVE score (changes) Column = PRINTED score (fixed)
Star vs Snowflake Schema	Star = Simple Snowflake = Complex (more branches)
Import vs DirectQuery	Import = Downloaded movie DirectQuery = Netflix streaming
Workspace Roles (AMCV)	Always Make Clear Visuals: Admin > Member > Contributor > Viewer
Row vs Filter Context	Row = one row at a time Filter = entire dataset with a filter
Gateway Analogy	Gateway = Postman delivering local data to the cloud
Slicer vs Filter	Slicer = visible on canvas (user controls) Filter = hidden in filter pane
Dataflow	Dataflow = Power Query's twin living in the cloud